

Speed Safety Score — what this tool shows

This calculator demonstrates the core analytical method behind a road safety assessment built for the Asian Development Bank's AI for Safer Roads Innovation Challenge. The question it answers is not whether drivers are speeding, but whether the posted speed limit is appropriate for the road in front of them.

Every road segment in the dataset receives a Speed Safety Score — a single number that combines three independently interpretable components.

Severity captures how dangerous the operating conditions are relative to what the road can safely support. It applies Nilsson's Power Model, a well-established relationship in road safety research: fatal crash risk scales with the fourth power of speed. A road where traffic moves at 65 km/h on a section benchmarked for 50 km/h does not carry 30% more fatal risk — it carries roughly 2.9 times the risk. The benchmark itself comes from Safe System principles: biomechanical research on what the human body can survive in different collision types.

Exposure converts the vulnerability of road users into a multiplier. It draws on helmet-wearing rates (collected as Safety Performance Indicators by the ADB) and the typical presence of motorcyclists, pedestrians, and cyclists on that type of road. A busy urban corridor with low helmet compliance amplifies the consequences of any crash; a rural motorway with high compliance does so less.

Consequence reflects how forgiving the road infrastructure is if a crash does occur. Roadsides free of rigid objects, roads with proper shoulders and sealed surfaces, and carriageways with physical median barriers all reduce fatality risk even at the same speed. This component is derived from Google Street View imagery analysed by a multimodal AI model (Gemini 2.5 Flash), which classifies each segment against an iRAP-inspired rubric covering roadside hazard, barrier type, shoulder condition, and surface quality.

The three components multiply together. This matters: a high-speed road with excellent infrastructure and protected users produces a different score than the same road with unprotected motorcyclists and hazardous roadsides — even if the raw speed readings are identical. Each component can be inspected independently, which makes the score defensible to policy audiences rather than a black box.

Interpreting the score

A score below 2 means the road is broadly aligned — operating speed, infrastructure, and user vulnerability are in reasonable balance. Scores between 2 and 5 flag moderate concern worth monitoring. Above 5 the mismatch is material; above 10 it is critical. The logarithmic spread of the scale reflects the exponential nature of the Power Model: scores at the high end represent multiplicative risk accumulation, not linear excess.

Intervention diagnosis

The tool also assigns each segment an intervention type. This uses two signals: Signal A measures whether the posted limit itself is too high relative to what the infrastructure

supports; Signal B measures whether drivers are exceeding even that limit. Combining these with the road's visual character — does the geometry invite high speed or constrain it? — produces four possible diagnoses. A limit reduction is warranted when the posted number is simply too generous for the road class. Traffic calming is needed when the environment feels like a fast road and drivers behave accordingly, regardless of what the signs say.

Enforcement addresses non-compliance on roads that already signal appropriate speeds.

An enforcement crisis flags the rare and serious case where the limit is already conservative but compliance has collapsed. Each diagnosis also carries an expected fatality reduction estimate, calculated using the same Power Model that drives the Severity component.